	Document No.	Section No.
	Effective Date:	Revision No.
	PROCEDURAL MANUAL	
	EROSION AND SEDIMENT CONTROL	

OBJECTIVE	This procedure outlines the requirements for planning, implementing, inspecting, and maintaining erosion and sediment control (ESC) measures on BHBI construction sites. The aim is to minimize soil erosion caused by construction activities, prevent sediment and other pollutants from leaving the site and entering waterways or adjacent properties, protect water quality, and ensure compliance with relevant environmental regulations (including DENR requirements and local ordinances)
SCOPE	This procedure applies to all BHBI construction activities involving land disturbance, including but not limited to: • Site clearing and grubbing • Grading and excavation • Trenching • Stockpiling of soil and materials • Construction site access and egress • Dewatering operations

Responsibilities

- **Project Manager:** Overall responsibility for ensuring project compliance with ESC regulations and requirements.
- **Project Engineer:** Responsible for developing or overseeing the development of the site-specific Erosion and Sediment Control Plan (ESCP), ensuring adequate resources are allocated for ESC implementation, and overall compliance.
- **Site Engineer:** Responsible for the day-to-day implementation, inspection, and maintenance of ESC measures on site according to the approved ESCP. Coordinates ESC activities with Foremen and subcontractors.
- **Safety Officer:** Assists the Site Engineer in monitoring ESC measures, ensuring they do not create safety hazards, and integrating ESC awareness into safety briefings.
- **Foremen:** Responsible for ensuring their crews implement and maintain assigned ESC measures within their work areas and report any damage or issues to the Site Engineer.
- **Equipment Operators:** Responsible for operating equipment in a manner that minimizes land disturbance and avoids damage to ESC measures.
- **Subcontractors:** Responsible for implementing and maintaining ESC measures related to their specific scope of work, as outlined in the ESCP and their contract.
- **All Site Personnel:** Responsible for being aware of ESC measures and avoiding actions that could compromise their effectiveness.

Procedure

4.1. Planning and Design (ESCP):

- **Develop ESCP:** Prior to commencing significant land-disturbing activities, the Project Engineer, with assistance from the Site Engineer, develops a site-specific Erosion and Sediment Control Plan (ESCP). The ESCP considers site topography, soil types, drainage patterns, proximity to water bodies, project phasing, and local rainfall patterns. (Refer to *Mobilization and Pre-Project Activities Procedure*).
- **ESCP Content:** The ESCP typically includes:
    - Site map showing disturbed areas, slopes, drainage paths, receiving waters/properties.
    - Locations and details of planned ESC Best Management Practices (BMPs).
    - Construction phasing and sequencing related to ESC implementation.
    - Inspection and maintenance schedule and procedures.
    - Spill prevention measures related to potential soil/water contaminants.
    - Final stabilization plan.
- **Regulatory Compliance:** The ESCP must comply with applicable DENR regulations (e.g., conditions of Environmental Compliance Certificate - ECC, if applicable; relevant DAO provisions) and local government ordinances.
- **BMP Selection:** Select appropriate ESC BMPs based on site conditions and regulatory requirements. Common BMPs include silt fences, check dams, sediment traps/basins, stabilized construction entrances, mulching, temporary seeding, erosion control blankets, etc.
    - **Deliverables:** Site-Specific Erosion and Sediment Control Plan (ESCP)

4.2. BMP Implementation - Perimeter Controls:

- **Install First:** Install perimeter controls (e.g., silt fences, sediment barriers) *before* major land disturbance begins in an area to contain sediment runoff within the site boundary.
- **Discharge Points:** Install controls (e.g., check dams, sediment traps) at points where concentrated flow leaves the site or enters site drainage systems.

4.3. BMP Implementation - Erosion Control:

- **Minimize Disturbance:** Phase construction activities to minimize the extent and duration of soil exposure. Protect existing vegetation where possible.
- **Surface Roughening:** Roughen graded slopes to slow runoff and promote infiltration.
- **Temporary Stabilization:** Apply temporary stabilization measures (e.g., hydraulic mulch, temporary seeding, erosion control blankets) to disturbed areas that will be inactive for a specified period (e.g., 14 days) or as required by regulations/ESCP.
- **Permanent Stabilization:** Implement permanent stabilization (e.g., topsoiling, permanent seeding, landscaping, paving) as soon as practicable after final grading is complete.

4.4. BMP Implementation - Sediment Control:

- **Sediment Traps/Basins:** Install sediment traps or basins where concentrated runoff occurs to allow sediment to settle out before water is discharged. Design according to anticipated flow rates and required sediment removal efficiency.
- **Check Dams:** Install check dams (e.g., rock, fiber rolls) in ditches or swales to slow flow velocity and trap sediment.
- **Inlet Protection:** Protect storm drain inlets from sediment influx using appropriate filters or barriers.

4.5. BMP Implementation - Site Access/Tracking Control:

- **Stabilized Entrance/Exit:** Install and maintain a stabilized construction entrance/exit (e.g., coarse aggregate pad) to minimize tracking of mud onto public roads.
- **Street Sweeping/Cleaning:** Regularly sweep or clean public roads adjacent to the site entrance if tracking occurs.
- **Wheel Wash:** Implement wheel washing facilities if required by site conditions or regulations.

4.6. BMP Implementation - Stockpile Management:

- **Location:** Locate soil and material stockpiles away from drainage paths, water bodies, and steep slopes.
- **Containment:** Surround stockpiles with perimeter controls (e.g., silt fence).
- **Covering:** Cover inactive stockpiles (e.g., with tarps) to prevent erosion by wind and rain, especially before predicted heavy rainfall.

4.7. Inspection and Maintenance:

- **Regular Inspections:** The Site Engineer or designated personnel inspects all installed ESC BMPs regularly (e.g., at least weekly) and **before anticipated storm events, during storm events (if safe to do so), and within 24 hours after storm events exceeding a specified threshold (e.g., 12mm)**. This is critical given Philippine weather patterns.
- **Documentation:** Record inspection findings, including the condition of BMPs, any required maintenance, and corrective actions taken.
    - **Deliverables:** ESC Inspection Log
- **Prompt Maintenance:** Repair or replace damaged or ineffective ESC measures immediately upon discovery. Remove accumulated sediment from traps, basins, and behind barriers when they reach a specified capacity (e.g., 50% full) or lose effectiveness. Maintain stabilized entrances.

4.8. Dewatering Operations:

- **Sediment Removal:** Water pumped from excavations or low areas must be treated to remove sediment *before* being discharged from the site or into protected waterways.
- **Treatment Methods:** Route dewatering discharge through sediment traps, basins, or filter bags/tanks. Ensure discharge does not cause erosion at the outlet point.

4.9. Final Stabilization:

- **Requirement:** All disturbed areas must be finally stabilized (e.g., vegetation established, paving completed) upon completion of construction activities in that area.
- **Removal of Temporary BMPs:** Remove temporary ESC measures only after the area they are protecting is fully stabilized. Dispose of removed materials appropriately (*Waste Management Procedure*).

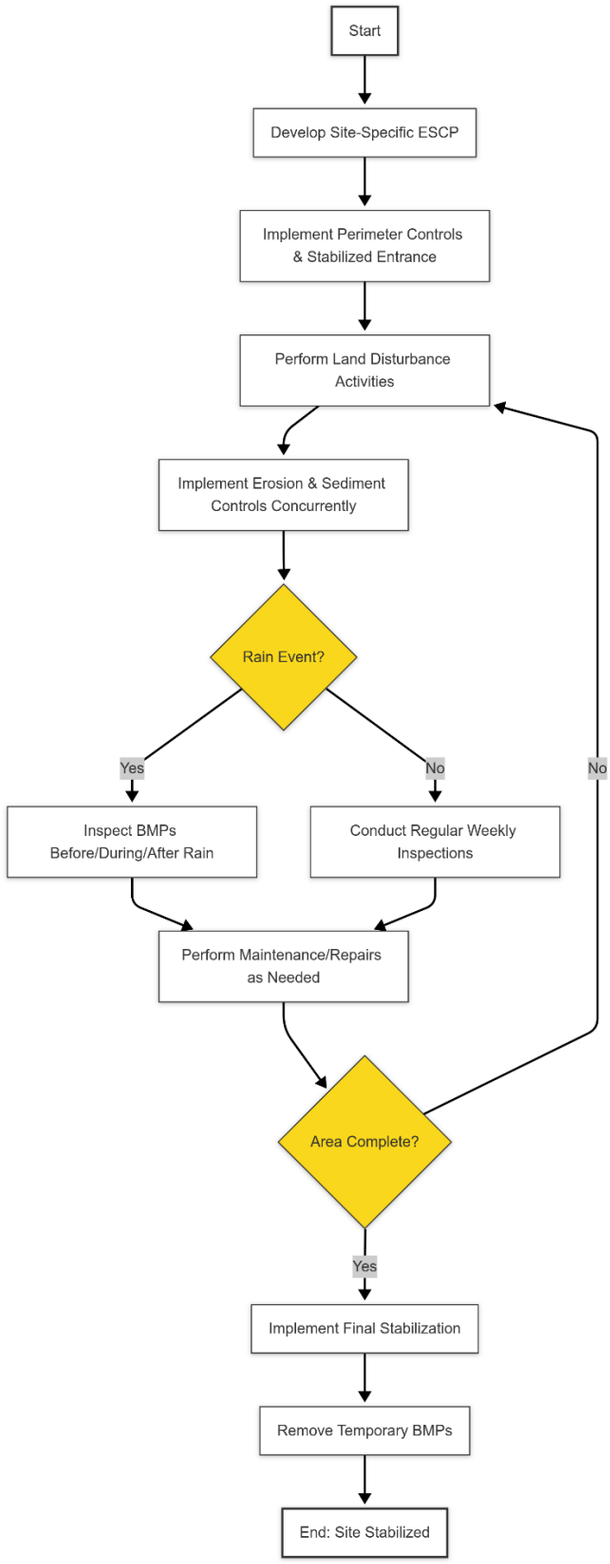
4.10. Training:

- **Induction & Toolbox Talks:** Include ESC awareness and requirements in site inductions and regular toolbox talks. Emphasize the importance of protecting installed BMPs and reporting issues.
- **Specific Training:** Provide specific training to personnel responsible for installing and maintaining ESC BMPs.
    - **Deliverables:** Training Records

4.11. Record Keeping:

- **Maintain Records:** The Project Engineer/Office Engineer maintains copies of the approved ESCP, permits, inspection logs, maintenance records, training records, and photos documenting ESC implementation.
    - **Deliverables:** ESCP, Permits (if applicable), ESC Inspection Logs, Maintenance Records, Training Records, Photographic Records

Flowchart



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